Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2419)

Key

SECTION – I

1.	A	7.	C	13.	D	19.	4213
2.	C	8.	D	14.	D	20.	2431
3.	B	9.	C	15.	A	21.	D
4.	D	10.	B	16.	C	22.	B
5.	C	11.	B	17.	2431	23.	C
6.	A	12.	A	18.	C	24.	A

SECTION – II

1.	C	6.	B	11.	103	16.	A
2.	A	7.	41	12.	78	17.	C
3.	C	8.	B	13.	79	18.	D
4.	B	9.	70	14.	B	19.	C
5.	B	10.	81	15.	B	20.	B

SECTION – III

1.	A	7.	D	13.	C	19.	21
2.	98	8.	D	14.	6.5	20.	B
3.	D	9.	A	15.	12500	21.	C
4.	D	10.	6	16.	B	22.	26
5.	C	11.	C	17.	701		
6.	D	12.	B	18.	A		

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 528

1. Option A: The passage starts with the discussion of ***hiring*** and recruitment of candidates and how the current procedures do not suffice. Then it moves on to discussing the tools provided by ***data science*** which provide a solution for the ***social bias*** existing in the industry. This might inadvertently lead to the propagation of those biases further. Collection of ***data*** has been discussed next as a major impediment followed by how the algorithms propagate ***regressive predictions*** (predictions based on data of the past). Hence, Option A is the answer.
Option B: While recruitment is the first step, the second is not just algorithms but those driven by data science. Further, while prejudice and bias are the idea discussed, the passage doesn't move on to sexism from there. Sexism is discussed much later in the passage but as an example where women receive fewer scores. Finally, technology is introduced much earlier in the passage while discussing data science and not at the end. Hence, Option B is not the answer.
Option C: The passage doesn't start with employees. It instead discusses 'hiring and recruitment'. Also, it doesn't jump to social bias. It first moves to data science which doesn't even make an appearance here. Hence, Option C is not the answer.
Option D: While job candidates and social bias are important starting points of the discussion, this option doesn't bring in technology/data science/science in the beginning. The passage discusses regressive predictions right after data collection. So science is not part of the logical sequence of ideas from data collection to regressive predictions. Also, the term science here is very generic and need not be specific to this passage. Hence, it is not an apt representation of the logical sequence of ideas discussed in the passage. Choice (A)

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2. Option A: This has not been discussed as part of this example. The potential of women and men is not the discussion point here. The algorithm is the main issue. Hence, Option A is not the answer.
Option B: This option was a symptom of the problem, not the main cause. Through this, it could be observed that women were being given lower scores. Why they were being lower scores? Because men were found to be better performers (obviously, because majority were men). Hence, Option B is not the answer, even though it is not a false statement.
Option C: There were two problems with Amazon's hiring algorithm. Firstly, it gave lower scores to women on certain attributes where women cannot score lower than men, for example, in participation of women's studies (obviously women will be the subject of such studies). Secondly, the algorithm believed that men are better performers in Amazon because of past results. Amazon's scrapping of the algorithm is an acceptance of this flaw. The number of top performers among men is acceptedly better only if Amazon agrees that more men were hired by them. If the numbers are equal, then men being top performers wouldn't have led to the conclusion that the algorithm was faulty. Option C is the answer.
Option D: Amazon scrapped the algorithm because it was throwing up lower scores for women, which was because men were the better performers, which, in turn, was because in the past, Amazon had more male employees than female. But we cannot say that the algorithm was not picking the 'best' women candidates. Hence, Option D is not the answer. Choice (C)

3. Option A: Longer commuting is correlated to higher attrition. This is statistically understood according to the passage (the passage mentions there is data). So, it is not the author's assumption that longer commute makes people want to quit their jobs. Option A is not the answer.
Option B: Consider this: '*Picking whom to hire on the basis of where they live most likely has an adverse impact on protected groups such as racial minorities.*' This can be true, only if the author assumes that those who come from racial minorities live together. Only in that case, their commuting times can be generalised to be high, thereby making them vulnerable to an adverse impact from the algorithms that use commuting time as a parameter. Hence, Option B is the answer.
Option C: Higher salaries may or may not offset the stress caused by the higher commuting time. The author doesn't bring salary into the equation and the data doesn't mention whether the correlation between commuting and attrition is linear, with no dependence on salary. So, it is a vague concept and hence, cannot be understood to be assumed by the author. Option C is not the answer.
Option D: While the author does mention that commuting time is a function of where they live which is a function of their income levels, it has not been assumed (or implied) that there is only one area where people of a certain income group live. They need not cluster in the same area, but in similar areas scattered around the city, for example. Option D is not the answer. Choice (B)

4. Option A: In the last line of the passage, the author suggests we let go of the cheap meal (data science-driven algorithms for hiring) altogether. The author doesn't stop at saying they are overhyped (if they are overhyped, it means we could still use them). Option A is not the answer.
Option B: Data science–driven algorithms are not effective because vendors are trying to do it the cheap way and so are the companies. However, the last line isn't a call for improvement or refinement. It doesn't have a positive tone that provides a solution. It simply asks us to let go of this option. Hence, Option B is not the answer.
Option C: While this is one of the points the author mentions in the passage, this is not what the author indicates or discusses in the last line of the passage. The last line is a much bigger idea which doesn't talk about one drawback of data science-driven algorithms. Hence, Option C is not the answer.
Option D: The author clearly asks us to ignore the cheap meal altogether in the last line of the passage. This reiterates what the author mentions in the first para of the passage that mentions that data science isn't yet a panacea for hiring problems. So, the author does believe that we need to stick to the older methods for now. Or that, these algorithms aren't ready to replace the existing tools (since the author mentions that we 'skip' them). Option D is the answer. Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 563

5. Actual cases of art fraud are exceptionally rare, although cases of "mistaken identity" are reasonably common.
Option A: Many leading artists were more than happy to create and sell additional versions within the studio, probably with even less personal involvement than before. All these possibilities present significant art historical and ethical problems in terms of identification and attribution, but perhaps raise important legal issues as well. This means that 'mistaken identity' could be the result of 'identification' problems, and, by implication could lead to historical problems. There is nothing in the context to rule out this possibility, so Choice A is inappropriate since it says 'the latter does not'.
Option B: A deliberate deception is clearly criminal. Creating a deliberate forgery for sale or attempting to pass off a fake or copy as an original is

deception, and a criminal offence in most if not all countries. So the first part of choice B is true. The second part of choice B is not correct as can be inferred from" ...reckless and unsupported exaggeration of an identification or attribution (***either of which might be open to legal challenge*** in the civil courts for compensation)". Hence choice B is not the answer.
Option C: While fraud shares with the wider interests the need to demonstrate deception (an obvious aspect of a fake), a successful prosecution will require in addition that the defendant be shown to be dishonest (that is, that ***the deception is intentional***), that there is harm as a consequence, and that the victim was actually deceived.... It may not be easy to define exactly the boundaries between reckless and unsupported exaggeration of an identification or attribution, and a deliberate deception which is clearly criminal. Choice C is the answer.
Option D: Actual cases of art fraud are exceptionally rare, although cases of "mistaken identity" are reasonably common. However, a deliberate deception is observed in the case of art fraud and not in a case of 'mistaken identity'. Hence, the latter does not involve any deception skills. Choice D is not the answer. Choice (C)

6. Option A: Many leading eighteenth-century portrait painters personally concentrated on the overall composition together with the face and perhaps hands, leaving the greater part of the canvas to be filled in by others ... Leading artists were more than happy to create and sell additional versions within the studio. ... All these possibilities present significant art historical and ethical problems in terms of identification and attribution. It may not be easy to define exactly the boundaries between a perhaps mistaken opinion made in good faith, reckless and unsupported exaggeration of an identification or attribution (either of which might be open to legal challenge in the civil courts for compensation), and a deliberate deception which is clearly criminal. ..... We are asked ***what difference it makes if a work is produced exactly like the genuine one***. Hence choice A is the answer.
Option B: It is stated that there are examples where a subsequent version, though substantially the same, may be regarded as the superior version. But Choice B does not pose a significant historical and ethical problem. Hence, it is not the answer.
Option C: Choice C has been provided as a reason for art fraud to be infrequently observed (para 1) but it is not the answer to the question.
Option D: *We are asked what difference it makes if a work is produced exactly like the genuine one.* But the fact that any work of art will be appreciated or not does not pose any historical and ethical problem. The question of appreciation of a work of art is not addressed in the passage. Choice D is not the answer. Choice (A)

7. (a): The serious question is whether knowledge of these differences in any way impinges upon our appreciation of a work whose structure underdetermines the difference between authenticity and trumpery, or whether it makes no difference. ***I think one cannot say in advance whether it makes a difference or not.*** Hence (a) is correct and is the answer.
(b): (b) is out of scope of the given passage.
(c): The distinction between genuine and fake must be established with reference to factors external to the works themselves – for example, with reference to their histories. So (c) is not true.
(d): The method of philosophical ***duplication is a powerful lever for lifting factors into consciousness*** which otherwise never would have been alive – ***presumptions upon which our attitude towards the world has always depended*** though we might not have realized their crucial role since it never had been challenged. Therefore (d) is correct.
(e): .... All these possibilities not only present significant historical and ethical problems in terms of identification and attribution, but perhaps raise important legal issues as well. So (e) is correct and is the answer.
So, (a), (d) and (e) are true. Choice (C)

8. Option A: Choice A is wrong because the author does not compare human beings and artworks. Choice A is not the main reason for the question.
Option B: The author mentions that artwork can be duplicated, but he talks about the duplication of human beings only in hypothetical terms, not as something that really can be done.
Option C: Choice C is out of scope. We cannot verify from the passage that the duplicate belongs to a different era and that all factors are improved upon in the duplicate. Some factors might also be missing when a duplicate is made.
Option D: Philosophical duplication is a powerful lever for lifting factors into consciousness which otherwise never would have been alive – presumptions upon which our attitude towards the world has always depended though we might not have realized their crucial role since it never had been challenged. These factors will always be logically external to the thing in question. Hence choice D is the answer. Choice (D)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 502

9. The author mentions at the beginning of the second paragraph that "most business analysts remained tied to the past, wedded to a GM-era kind of thinking. At no time was this more evident than when Starbucks itself started to falter."
Option A: In the second paragraph, the author mentions that the business analysts stuck with old school economic reasoning to explain Starbucks' reversal of fortunes. This is why the author

mentions that they "remained tied to the past". The author does not mention that they failed to notice that the American economy moved in tandem with latte sales to support his claim. Hence, this is not the correct answer.
Option B: According to the author, "the experts had it largely wrong, in terms of both the timing and the causes of Starbucks' decline. That's because they repeatedly fell back on culturally uninformed, old school economic reasoning to explain Starbucks' slip." Hence, the author mentions that they used traditional economic reasoning to explain Starbucks' slip and not the slowdown of American economy.
Option C: The experts attributed Starbucks' slip to "rising price of gasoline, competition from McDonald's and Dunkin' Donuts, the mortgage crisis, and a new frugality bred by rising joblessness", which was a mistake according to the author. Because they stuck with traditional economic reasoning, the author says that they remained in the past. Hence, this is the correct answer.
Option D: In the second paragraph, where the author claimed that "most business analysts remained tied to the past", the author does not talk about production, jobs and purchasing. He chiefly talks about the reversal of Starbucks' fortunes. Hence, this is not the correct answer.
Therefore, the correct answer is option C. Choice (C)

10. The author states that "During GM's reign as the nation's financial bellwether, business in the United States revolved around production, employment, and consumption". Further, "During the days that the nation moved in tandem with Starbucks and latte sales, the American economy turned almost entirely on buying alone".
Option A: The author mentions that American economy moved in tandem with both these companies at different points in time. However, moving in tandem does not imply that these two companies determined the direction in which American economy moved.
Option B: During the GM era, "business in the United States revolved around production, employment, and consumption". GM was a major player in these aspects. In the Starbucks era, "buying drove the nation's economic engine". In the second paragraph, the author explains how Starbucks is symbolic of buying in United States. Hence, both the companies symbolized what the American economy depended on. This is the answer.
Option C: While both the companies acted as important indicators of the economy, we cannot infer that they were significant contributors to the growth of American economy. Hence, this is not the correct answer.
Option D: According to the author, when the nation moved in tandem with Starbucks, buying "shaped the daily lives identities, and emotions of the country's citizenry". However, he does not say the same about GM era. Further, he mentions in the second paragraph that the factors which people consider important in buying in the GM era were no longer as important in the post-GM era. Hence, this is not the correct answer. Choice (B)

11. The company talks about the relation between Starbucks and the American economy in the second paragraph of the passage.
Option A: The author mentions that the experts were mistaken in thinking that "the rising price of gasoline, competition from McDonald's and Dunkin' Donuts, the mortgage crisis, and a new frugality bred by rising joblessness" was the reason for Starbucks' reversal of fortune. Hence, competition from other companies was not the reason, according to the author.
Option B: The author states that the experts "expected buying decisions to revolve around utility, cost, and the physical qualities of a product but, as Starbucks' spectacular success had demonstrated, buying in post-GM, postindustrial America turned on more than price or functionality." The experts believed that utility, cost and physical qualities of a product impact buying decisions. But they were proved wrong because of Starbucks' success. The price (or affordability) of Starbucks' products was not the only reason for its success. There were other reasons (like an expression of longing, a source of entertainment...) for its success. Hence, this is the correct answer.
Option C: The author mentions that Starbucks enjoyed "a fifteen-year uninterrupted run of nonstop growth". But we do not know how the author defines growth in this case. However, we can infer that the author is talking about the number of stores in U.S. from the next part of the sentence. Therefore, we cannot infer anything about the stock price of Starbucks from this.
Option D: The author mentions that Starbucks became successful because of the change in the buying behaviour of the people. However, we cannot infer that Starbucks was responsible for this change in the people. Hence, this is not the correct answer.
Therefore, the correct answer is option B. Choice (B)

12. The author talks about a fundamental transformation in the last paragraph of the passage: "the nearly wholesale replacement of civic society by a rapacious consumer society." The author follows this by saying what was not new – "hefty doses of buying, advertising, and marketing", "the branding of everything"... What's new is mentioned in the last sentence of the paragraph and this can be inferred to be a feature of a "rapacious consumer society".
Option A: According to the author, the change was "public institutions that in the past had pushed back against the market and brands to challenge them for people's allegiances and identities". Hence, the public institutions which in the past

challenged markets and brands for people's allegiances are no longer strong. The public institutions do not stand between the markets and the people. Hence, this is the correct answer.
Option B: The author mentions at the beginning about the "**nearly** wholesale replacement of civic society by a rapacious consumer society". Since the author said that the civic society is nearly replaced, we cannot say that the markets and brands have taken over all the functions of public institutions.
Option C: We cannot infer from the passage that the public institutions and civic society need branding for functioning effectively as the passage does not talk about effective functioning of civic society.
Option D: The author does not talk about effective functioning of civic society and markets. Hence, this cannot be inferred from the passage.
Therefore, the correct answer is option A. Choice (A)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 562

13. To start with the reasons Einstein could not have foreseen here make lambda a possibility. Lambda is a possibility because something needs to explain the acceleration of the expansion of the universe, something that counters the gravity. So, the 'reason' here is the acceleration, which the author feels Einstein couldn't have foreseen.
Option A: A small error in the calculation of the brightness of the supernova will distort other calculations. That is true. However, that doesn't really contradict the reason why the universe is expanding. Since, this option doesn't prove that there was an actual error in the calculation of the brightness of the supernova, this option doesn't really affect the argument. Option A is not the answer.
Option B: While this is true, it only strengthens the argument about the universe expanding faster and faster. It doesn't weaken that argument. Hence, Option B is not the answer.
Option C: This is known, and it doesn't affect the final argument. The author doesn't say that Einstein has definitely been proven to be right. The lambda maybe a possible candidate to explain the acceleration of the universe's expansion. So, the mysterious repulsion not being proven doesn't weaken the information about the accelerating speed of the expansion of the universe. Hence, Option C is not the answer.
Option D: If this is true, it weakens the reason why Einstein could be right. It means the basis on which researchers feel the expansion of the universe is accelerating could fall flat. Their calculations are based on the intrinsic brightness of the supernova. So, if we are wrong about the brightness, the rest of the calculations could be wrong. Hence, Option D is the answer. Choice (D)

14. Option A: The age of the Cepheid Variables/when they have been formed or other such data hasn't been mentioned in the passage. So, Option A is not the answer.
Option B: While this is indeed true, it is a circular argument, in that it doesn't tell us why Cepheid Variables cannot help us prove the slowing of the universe. Yes, we need light from billions of light years away, but why can't that light be from Cepheid Variables? This option doesn't explain why not. Option B is not the answer.
Option C: Cepheid Variables aren't at a fixed distance from us. They are present at increasing distances as can be understood from 'as more distant Cepheids appear dimmer'. So, this cannot be the reason. Option C is not the answer.
Option D: While Cepheid Variables may be present as far away as billions of light years, they aren't bright enough for us to spot them from such a distance. In that case, measurements of the expansion or rate of expansion of the universe cannot be made. Hence, Option D is the answer. Choice (D)

15. Option A: This can be understood from the following lines: 'But gauging a galaxy's actual distance is difficult. Hubble managed it by observing the apparent brightness of stars called Cepheid variables, whose intrinsic brightness is known.' So, the perceived brightness of Cepheid variables (as compared to their known brightness) helped us understand the actual distance of the galaxies. Option A is the answer.
Option B: The information that galaxies are moving away is not obtained from Cepheid variables. It is obtained from the 'redshift' as understood in the following lines: 'From our vantage point in the Milky Way, the speed at which any other galaxy is moving away can be clocked using the "redshift" of its light—a drop in frequency and increase in wavelength akin to the dip in pitch of a receding train's whistle.' Cepheid variables help us know about the distance of the galaxies. Hence, Option B is not the answer.
Option C: Einstein used the lambda to prevent the universe collapsing under the gravitational pull of the matter inside it. However, Einstein didn't prove the existence of lambda, and in fact, called it a blunder later. So, Option C is not the answer.
Option D: It is not just the dimming that proved the expansion of the universe. The dimming only gives us the distance of the galaxy. It is the redshift that proves that the galaxy is speeding further away, thereby, establishing the expansion of the universe. So, Option D is not the answer. Choice (A)

16. Option A: While Einstein indeed agreed with Hubble's discovery that the universe was expanding, that is the not the reason why he reasoned that the lambda isn't necessary, as understood from this sentence: *But he reasoned that if the expansion was a relic of a primeval explosion, the cosmological constant... wasn't needed.* Option A is not the answer.

Option B: While Einstein mentioned this, mere aesthetics cannot be the reason for him to remove the constant. It was just a side note and not the reason for getting rid of the constant as understood from the fact that this information was given between the two dashes. Option B is not the answer.
Option C: Einstein reasoned that ***the lambda may not be necessary if the expansion was a relic of a primeval explosion.*** In other words, if the universe was expanding because of an earlier explosion, the lambda, a repulsion energy to counter gravitational pull, isn't required in his equations. Option C is the answer.
Option D: This is not relevant to Einstein's reasons for withdrawing lambda, as the discussion about expansion and whether there is enough gravitational pull to counter it comes much later (in the passage and in the chronology of events discussed in the passage). Option D is not the answer. Choice (C)

Solution for question 17:

17. Sentence 1 talks about 'these' commentaries (based on another sentence which introduces such commentaries) and the role they play in preserving Greek works.
Sentence 2 talks about a person, obviously, whose father was Theon. But it is an independent sentence that establishes the context about possibly someone who was into math and astronomy.
Sentence 3 talks about Hypatia and how texts written by her were attributed to her father. Please note that the 'her' in the sentence doesn't make this sentence independent, as the name does pop up later in the sentence.
Sentence 4 talks about how Hypatia helped Theon with his commentaries.
The meaning of commentaries as republications has been mentioned in 4. So, 4 precedes the other sentence that uses 'commentaries', 1 and 3. 1 says 'commentaries such as these' and the sentence with an example of a commentary is in 3. So, 31 is a block, preceded by 4. Since, 2 establishes the context by introducing Theon as the father, it comes first. Hence, the sequence is 2431. Ans: (2431)

Solution for question 18:

18. Option A: Though choice A might seem implied, it is not the main difference between the two societies. The author clearly mentions that individualistic societies encourage children to make personal choices early on in their lives whereas collectivist societies stress upon duty rather than personal choice.
Option B: Indian children have not been spoken about in the passage. Besides, it has been stated that the elders decide for the children in collectivist societies. But it has not been stated that these children *prefer* to let their elders decide for them. So choice B is incorrect.
Option C: Choice C is a complete, concise and correct summary. Choice C is the answer.
Option D: In choice D, the point about "individualistic societies" is correct. The remaining part of the choice merely repeats some examples (which need not be mentioned in the summary). Also "should be determined by society and should not be left to the whims of the individual" in choice D is out of scope. Choice D does not highlight the fact that collectivist societies emphasize duty. Hence choice D is not the answer. Choice (C)

Solutions for questions 19 and 20:

19. Sentence 1 talks about 'greening' in urban policy. It is an independent sentence.
Sentence 2 starts with a contrast marker 'but' followed by things brewing (changing, new things happening) in cities.
Sentence 3 starts with 'This gathering of energies' referencing 1 which talks about 'gathering some momentum'. So, 13 is a block, both together pointing to policies investing in 'urban green'.
Sentence 4 is an independent sentence that talks about how urban ecology was a concept that wasn't seemingly practical – a contradiction.
Since 13 talks about urban greening and 4 talks about the contrary, they are on either side of 'But' in 2. Also, since the right side of 'but' here is pointing to the present, things changing, 4 has to precede 2, whereas 13 will follow 2. Ans: (4213)

20. Sentence 1 talks about some difference discussed in a preceding sentence.
Sentence 2 is an independent sentence establishing a context – that tissue repair is important in the heart.
Sentence 3 discusses heart repair (downstream from Sentence 2) and the role of macrophages.
Sentence 4 also discusses heart tissue repair, in foetal mammals and most species which lose the ability as adults.
We can understand that the 'difference' mentioned in 1 is the difference between 3 and 4. Most species lose the ability as adults. But in mouse foetuses, macrophages play a starring role in scar-free repair, in adults. So, 43 is a block followed by 1. 3 comes second because in 4 the author says, 'scientists don't fully understand' which is a vaguer, bigger leading idea compared to 'researchers learning something'. So, the order is 2431. Ans: (2431)

Solution for question 21:

21. While the book mentioned in the highlighted sentence could refer to The Man in the High Castle, the witch hunts and communism aspects of the highlighted sentence cannot be ascribed to the book as these things have been touched upon in the later part of the paragraph. Also, the phrase "the book showed" should follow a statement that

discusses how it was shown, which is not mentioned in the statement preceding blank (1). Hence, the highlighted sentence does not fit in blank (1).
The highlighted sentence is more of a conclusion as to what the book entailed. It needs a precedence that explains Dick's politics. The sentence preceding blank (2) only mentions the book Eye in the Sky. Hence, the highlighted sentence does not fit in blank (2).
The statements immediately preceding and following blank (3) talk about the aspects of the story but the highlighted sentence is a general idea about the book. Hence, the highlighted sentence does not fit in blank (3).
The statements preceding blank (4) talk about what the book is about and how some of its characters have communist leanings. It highlights Dick's stories having a political dimension, which is mentioned at the beginning of the paragraph. The over-the-top Marxist world conjured by a communist is something that could be alluded to Dick's work not having a straightforward description of the left or right. Therefore, it can be inferred that the highlighted sentence concludes the paragraph. Hence, the highlighted sentence fits in blank (4). Choice (D)

Solution for question 22:

22. Option A: While the para does hint about the subconscious, its essence is not to distinguish between what the artist is trying to tell consciously and what the artist is trying to tell subconsciously. Yes, artists let go of their own personality, but it hasn't been mentioned that true art is not about what the artist is consciously trying to narrate. The 'subconscious aspect of an artist's thought' is not a good representation of 'how it is said'. Option A is not the answer.
Option B: This represents both the ideas in the para – that true artists produce art that is not about superficial concepts that content is important or that art is relevant. True art goes beyond this, because true artists have let go, of their personality, or of the surface personality. Option B is the answer.
Option C: Art doesn't become a phantom or ghost because the artist has let go. The artist becomes a phantom by letting go – a true artist, who focuses on the art rather than the surface personality. The tone of this option is exactly opposite to what the para is trying to convey. Option C is not the answer.
Option D: While genuine art doesn't subscribe to a specific form, art itself doesn't evolve unconsciously. True artists let go of the surface personality, so the unconscious aspect of the art, how it is said rather than what is said, comes out. So, Option D is not the answer. Choice (B)

Solution for question 23:

23. The observations made in statements preceding and following blank (1) are specific to the current time period. The observation made in the highlighted sentence is across different time periods. Therefore, inserting the highlighted sentence in blank (1) would break the flow. Hence, the highlighted sentence does not fit in blank (1).
The statements immediately preceding and following blank (2) talk about the collective cognitive capability being visualized in the form of a graph. The highlighted sentence explains why the graph was so. "we will conclude that the trend line has a strictly positive slope" in the statement following blank (2) means that the cognitive capability is increasing with time. The highlighted sentence validates the conclusion based on the graph. Therefore, the highlighted statement should follow both these statements. Hence, the highlighted sentence does not fit in blank (2).
The statements preceding blank (3) illustrate the idea mentioned in the beginning of the paragraph in the form of a graph. "we will conclude that the trend line has a strictly positive slope" in the statement preceding blank (3) means that the cognitive capability is increasing with time. The highlighted sentence validates the conclusion based on the graph. Hence, the highlighted sentence fits in blank (3).
"this trend" in the statement following blank (4) refers to "life has become only smarter and smarter" mentioned in the statement preceding blank (4). Therefore, these two statements are directly connected and inserting the highlighted sentence in between would break the flow. Hence, the highlighted sentence does not fit in blank (4). Choice (C)

Solution for question 24:

24. Option A: This option represents the following ideas: social behaviour is resource-heavy, it is important for evolutionary fitness, and in some species the adaptation to use biological mechanisms from elsewhere to address this more critical requirement has been witnessed. This option captures all the main ideas of the para. It is the answer.
Option B: Firstly, the para gives humans and monkeys as examples of some of the species which divert resources for social behaviour. It is not just monkeys and humans, necessarily. Also, the para doesn't mention why resources are diverted for the social behaviour-related cognition. Option B is not the answer.
Option C: This para does give us the main idea, but doesn't really tell us how the dependence on mental machinery has been addressed, missing out on the essence of the second half of the para. Hence, Option C is partially true, but doesn't represent the essence of the para. Option C is not the answer.

Option D: The causal relationship established in this option is not true. We don't need to observe humans, for example, to understand the importance of social information. The essence of the para is also missed out in the fact that the option focuses on humans and monkeys directly and also doesn't explain how the biological mechanisms have been rewired. Option D is not the answer. Choice (A)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	1, 10, 14, 16, 18, 23, 24
Difficult	2, 3, 4, 5, 6, 9, 11, 12, 13, 15, 17, 19, 20, 21, 22
Very Difficult	7, 8

SECTION – II

Solutions for questions 1 to 5:

Given that the monthly salaries are \$347, \$476, \$538, \$627, \$836, and \$968. The annual incomes, if no bonus was paid, will be monthly salary × 12, i.e., \$4164, \$5712, \$6456, \$10032, \$7524 and \$11616.

To make these annual incomes an integral multiple of \$1000 and ensuring that each employee was paid the least possible bonus, the employees will be paid a bonus of \$836, \$288, \$544, \$476, \$968 and \$384 (in the same order of the salaries mentioned above). The following table provides the salaries and bonuses of the six employees:

Salary	Bonus	Annual Income
347	836	5000
476	288	6000
538	544	7000
627	476	8000
836	968	11000
968	384	12000

From (i), the bonus paid to Dusk should be the same as the monthly salary of Busk. Hence, Dusk can be paid a bonus of \$836 or \$476 or \$968. The monthly salary of Busk must also be one of these three values.
The bonus paid to Gusk and the monthly salary of Fusk must also be \$836 or \$476 or \$968.
The least difference between monthly salary and bonus was for the employee with \$538 as the monthly salary. This cannot be any of Gusk or Fusk or Busk or Dusk. Also, from (ii), this cannot be Lusk. Hence, this must be Husk.
Busk can have a monthly salary of \$836, \$476 or \$968. His bonus in each of these three cases will be \$968, \$288 or \$384.
For Busk's monthly income in September (i.e., monthly salary plus bonus) to be less than the monthly salary of any employee, Busk's monthly income must be \$476, with his monthly income in September being \$764 (if his monthly salary is \$836 or \$968, his monthly income in September will not be less than anyone's monthly salary). This cannot be less than the monthly salary of Husk (whose monthly salary is \$538). Hence, it has to be less than the monthly salary of Gusk. Since Busk's monthly salary is \$476, Dusk's bonus should also be \$476. Hence, Dusk's monthly income must be \$627.
Hence, Gusk's monthly salary must be \$836 or \$968. However, Gusk's monthly salary cannot be \$968 as his bonus will be \$384, which cannot be the monthly salary of Fusk (from (i)). Hence, Gusk's monthly salary must be \$836 and his bonus must be \$968. Fusk's monthly salary must be \$968. Lusk's monthly salary must be \$347.
The following table provides the monthly salaries, bonuses and annual incomes of all the six employees:

Employee	Salary	Bonus	Annual Income
Lusk	347	836	5000
Busk	476	288	6000
Husk	538	544	7000
Dusk	627	476	8000
Gusk	836	968	11000
Fusk	968	384	12000

1. The annual income of Gusk was \$11000. Choice (C)

2. The bonus paid to Dusk is \$476.
The monthly salary of Fusk is \$968.
Required difference = \$492. Choice (A)

3. The bonus paid to Lusk was the same as the monthly salary of Gusk. Choice (C)

4. The monthly incomes for September for the six persons are \$1183, \$764, \$1082, \$1103, \$1804 and \$1352. Of these values, only \$1804 is a multiple of 11. Hence, only for Gusk the given condition is satisfied. Choice (B)

5. The monthly income for September as a proportion of annual income for
Husk = 1082/7000 = 15.46%
Lusk = 1183/5000 = 23.66%
Dusk = 1103/8000 = 13.79%
Fusk = 1352/12000 = 11.27%
It is the highest for Lusk. Choice (B)

Solutions for questions 6 to 10:

We can get the marks and the number of students who had a higher score than each of the three persons in different subjects from the graph. The table below provides, for each subject, the marks scored by each of the three students and the number of students who scored higher than each of the three students. In the score column, the name of the student (indicated by the first letter of the name) is provided in parenthesis along each mark.

Maths		Physics		Chemistry		Biology	
Score	No. of Students	Score	No. of Students	Score	No. of Students	Score	No. of Students
50 (K)	10	60 (L)	10	70 (L)	15	60 (R)	10
40 (R)	15	40 (K)	15	50 (R)	25	40 (L)	30
20 (L)	40	30 (R)	20	30 (K)	40	30 (K)	35

6. In Physics, Ram scored 30 marks and 20 students scored better than Ram.
Lalit scored 60 marks and 10 students scored better than Lalit.
Hence, among the 20 students who scored better than Ram, 10 must have scored better than Lalit, i.e., greater than 60 marks.
Hence, 20 – 10 = 10 students must have scored at most 60 marks. Since Lalit scored exactly 60 marks, 9 students must have scored less than 60 in Physics. Choice (B)

7. Since in Maths and Chemistry, there are 40 students who scored better than Lalit and Kishore respectively, the number of students in the class is a minimum of 40 + 1 = 41 (including Lalit or Kishore). Ans: (41)

8. For the given condition to be satisfied, the difference in marks between any two students must be less than or equal to the number of students whose marks are between the two marks.
For Maths, number of students who scored more than Ram is 15 and number of students who scored more than Lalit is 40. The number of students whose marks are less than Ram's and greater than Lalit's is 40 – 15 – 1= 24.
The number of positive integers between 40 and 20 (Lalit's and Ram's marks in Maths) = 40 – 20 – 1 = 19.
Since each student received a different mark, the given condition is not possible for Maths, as there will only be 19 distinct marks and 24 students.
For Physics, the difference in marks between Ram and Kishore = 40 – 30 – 1= 9, which is greater than the number of students, which is equal to 20 – 15 – 1 = 4.
Also, the difference in marks between Kishore and Lalit (= 19) is also greater than the number of students (= 4).
Hence, Physics can be one of the subjects.
Checking the differences for Chemistry, we can see that the difference between marks for any pair of persons is greater than the number of students.
For Biology, the number of students who scored more than 40 are 30 and the number of students who scored more than 60 are 10. Hence, there must be 20 students who scored more than 40 and at most 60. If these 20 students, one student is Ram, who scored 60 in Biology. Hence, there must be 19 more students who scored more than 40 and at most 59 (19 distinct integers). Hence, Biology can also satisfy the given condition.
Hence, for three subjects, Physics, Chemistry and Biology, the given condition holds.
∴ The maximum value of n = 3. Choice (B)

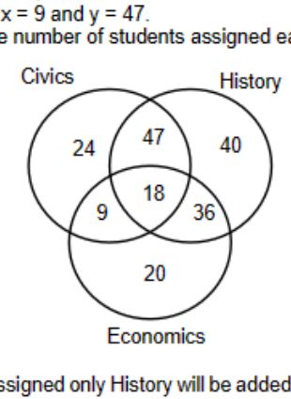
9. From the above solution, we can see that the marks scored by the students in Maths, Chemistry and Biology can be positive integers.
For Maths, 40 students scored more than Lalit, who scored 20 marks. Hence, there can be a maximum of 19 students who scored less than Lalit (since there are 19 positive integers less than 20).
Total number of students = 40 + 19 + 1 (including Lalit) = 60.
For Chemistry, 40 students scored more than Kishore, who scored 30 marks. Hence, there can be a maximum of 29 students who scored less than Kishore.
Total number of students = 40 + 29 + 1 = 70.

For Biology, 35 students scored more than Kishore, who scored 30 marks. Hence, there can be a maximum of 29 students who scored less than Kishore.
Total number of students = 35 + 30 = 65.
Since it is given that in only one subject, the students scored marks which are positive integers, to maximize the number of students, we can assume this subject to be Chemistry.
Hence, the maximum number of students in the class = 70. Ans: (70)

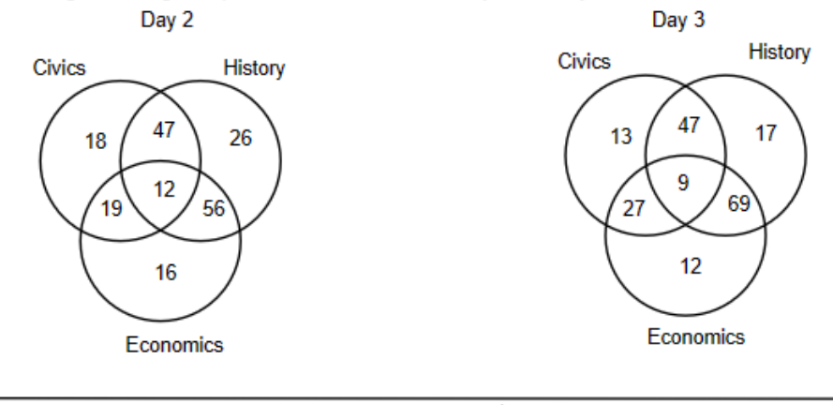
10. The maximum number of students who scored more than Lalit = 40 for Maths.
We can consider this to be the number of students who scored less than Kishore.
To maximize the number of students in the class, we can assume that the number of students who scored less than Kishore in Chemistry to be 40.
Since 40 students scored more than Kishore in Chemistry, total number of students in the class = 40 + 40 + 1 = 81. Ans: (81)

Solutions for questions 11 to 15:

We can start by solving for the numbers of students who were assigned the three subjects on Day 1.
Let the number of students who were assigned only Civics and Economics be x. The number of students who were assigned all three subjects will be 2x. The number of students who were assigned only History and Economics will be 4x.
Let the number of students who were assigned only Civics and History be y.
Given that the number of students who were assigned History = 141.
Since 40 students were assigned only History, 40 + 2x + 4x + y = 141 ⇒ 6x + y = 101
Since there are a total of 194 students, 40 + 24 + 20 + 4x + 2x + x + y = 194 ⇒ 7x + y = 110
Solving the two equations, we get x = 9 and y = 47.
The following diagram provides the number of students assigned each subject on Day 1:



On Day 2, 20 students who were assigned only History will be added to the group of 36 students who were assigned only History and Economics, making this 56.
10 students who were assigned only Economics will be added to the group of 9 students who were assigned only Civics and Economics, making this 56.
12 students who were assigned only Civics will be added to the group of students who were assigned all the three subjects. Also, the 18 students who were assigned all the three subjects will be equally distributed (6 each) to the groups of students who were assigned only Civics, only History and only Economics.
Similarly, we can calculate the number of students who were assigned the three subjects on Day 3 as well.
The following Venn diagrams provide the distribution on Day 2 and Day 3:



11. Number of students who were assigned Economics on Day 2 = 19 + 12 + 56 + 16 = 103 Ans: (103)

12. The number of students who were assigned Economics and History on Day 3 = 9 + 69 = 78 Ans: (78)

13. The number of students who were assigned only History on Day 3 = 17
The number of students who were assigned Civics on Day 2 = 18 + 47 + 19 + 12 = 96
Required difference = 96 – 17 = 79 Ans: (79)

14. On both Day 2 and Day 3, the number of students who were assigned Civics was less than that of those who were assigned Economics. Choice (B)

15. On Day 1, 40 students were assigned only History. On Day 2, 20 of these students were assigned one other subject. Hence, there are 20 students who were still assigned only History. Also, there were 6 students who were assigned only History on Day 2 (but not on Day 1).
On Day 3, 13 of these students were assigned another subject. Of these 13 students, 6 students can be the ones who were assigned only History on Day 2 but not on Day 1. The remaining 7 students must be from the 20 students who were assigned only History on Day 1 and Day 2.
Hence, at most 13 students could have been assigned only History on each of the three days. Choice (B)

Solutions for questions 16 to 20:

Given that the score of the player increases by 1 every time he hits the target. To maximize his score, he needs to increase his winning streak. But if he wins in consecutive rounds, he has to lose in consecutive rounds i.e., his losing streak will also increase. Hence, the points that he loses will also increase.

So, to maximize the points, he has to hit in consecutive rounds and distribute the misses so that the misses are not in consecutive rounds.

16. Given that n = 10 and the player missed in exactly three rounds. Hence, he hit the target in 7 rounds. If he hit the target in 7 rounds, the maximum score that he can get from these seven rounds is 3 + 4 + 5 + 6 + 7 + 8 + 9 = 42.
The maximum points that he can lose by missing 3 rounds = 1 + 2 + 3 = 6.
Hence, if he won continuously and lost continuously, his minimum score = 42 – 6 = 36.
However, we can minimize the score by breaking the series of wins into 4 different series by inserting three misses in between these hits.
For example, if he missed the 3rd, 6th and 9th round, the total points that he scored will be 3 + 4 – 1 + 3 + 4 – 1 + 3 + 4 – 1 + 3 = 21.
We can try shifting the -1's around to try minimize the score further. However, by trail and error, we can see that placing the -1's in any place will keep the score constant or will increase the score.
Hence, the minimum possible score of the player = 21. Choice (A)

17. Given that n = 10 and the player hit in exactly two rounds. Hence, he missed in 8 rounds.
To minimize the score of the player, we need to increase the points deducted.
If he missed in 8 consecutive rounds, the points deducted = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36.
We also need to minimize the score from the hits. If he hit in 2 consecutive rounds, his score will be 3 + 4 = 7.
However, if he hit in the first and last round, his score will be 3 + 3 = 6.
Hence, the minimum score can be obtained if he hit the target in the first and last rounds and missed the target in all the other rounds.
Total points = 6 – 36 = –30. Choice (C)

18. Given that the player scored 26 points in 10 rounds.
From the above solution, we can see that if he hit the target in 7 rounds, the maximum that he can score is if he missed the first two rounds and the last round.
The maximum possible score = 42 – 1 – 1 – 2 = 38.
If he missed in 4 rounds, the maximum possible score = 33 – 1 – 2 – 1 – 2 = 27 (missing the target in the first two and last two rounds).
If he missed in 5 rounds, the maximum possible score = 25 – 1 – 2 – 3 – 1 – 2 = 16 (missing the target in the first three and last two rounds).
Hence, if he scored 26 points, the maximum number of rounds in which he can miss is 4 (by missing the 1st, 2nd, 3rd and 10th rounds). Choice (D)

19. Given that n = 60 and the player hit the target in four rounds.
Hence, he missed the target in 56 rounds.
If he misses in continuous rounds, the points deducted = 1 + 2 + 3 + ... 56 = $\frac{56\times57}{2}$ = 1596.
However, we need to break this continuous series into smaller number of series.
With four hits, we can break this series of 56 numbers into five parts (which will be separated by one hit breaking the losing streak).
Making them as equal as possible, we will have four series each from 1 to 11 and one series from 1 to 12.
Points deducted in this case = 4 × (1 + 2 + 3 + ... 11) + (1 + 2 + 3 + ... 12)
$= 4 \times \frac{11\times12}{2} + \frac{12\times13}{2} = 264 + 78 = 342.$
Points awarded for hitting the target = 3 × 4 = 12 (since none of the hits were in consecutive rounds).
∴ The maximum possible score of the player = 12 – 342 = –330. Choice (C)

20. Given that n = 95 and he hit the target in 5 rounds. Hence, he must have missed the target in 90 rounds.
If he missed the target in 90 consecutive rounds, points deducted will be 1 + 2 + ... 90.
To maximize the number of points, we can split this series of 90 numbers into six series of continuous misses.
We can have six rounds of 1 to 15 with the five hits breaking the losing streak.
In this case, he will hit the target in the 16th round, 32nd round, 48th round, 64th round and 80th round.
The points deducted = 6 × (1 + 2 + ... 15)
If we shift any hit by even one round, the points deducted will increase. We can check this by assuming that he hit in the 33rd round, instead of the 32nd round.
Points deducted in this case = (1 + 2 + ... 15) + (1 + 2 + ... 16) + (1 + 2 + ... 14) + 3 × (1 + 2 + ... 15)
Comparing this with the above sum, we can see that we have replaced one 15 with on 16, increasing the overall points deducted.
Hence, to maximize the score, the player must have lost in the 16th round, 32nd round, 48th round, 64th round and 80th round.
From the given options, the answer is option B. Choice (B)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	1, 2, 3, 4, 5, 6, 7
Medium	8, 9, 10, 11, 12, 14, 16, 17
Difficult	13, 15, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. When the hypotenuse is reduced by 35% by cutting to along a line parallel to the hypotenuse, the perpendicular sides also get reduced by 35%. That is, the entire triangle is similar to the original triangle.
∴ Each side is now 65% of its initial value.
∴ New area is 65% × 65% of the area of the initial triangle.
= 42.25% of original area (since A = $\frac{1}{2}ab$ ⇒ A ∝ ab)
⇒ x = 42 (rounded off to the nearest whole number) Choice (A)

2. $f(n) = 1(1!) + 2(2!) + 3(3!) + ... + n(n!)$
$= (2–1)\,1! + (3–1)\,2! + (4–1)\,3! + ... + [(n+1) – 1]n!$
$= 2! – 1! + 3! – 2! + 4! – 3! + ... + (n + 1)! – n!$
$= (n + 1)! – 1$
∴ $f(117) = 118! – 1$ and $f(111) = 112! – 1$
∴ $f(117) + f(111) = 118! + 112! – 2$
As both 118! And 112! are multiples of 100, the remainder is 100 – 2 = 98.

Alternative Solution:

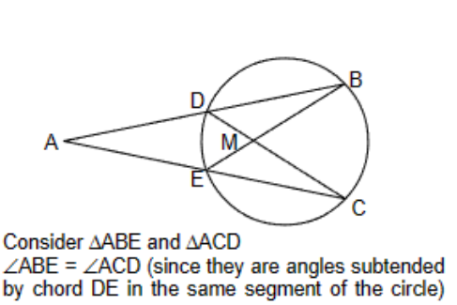
Since f (10) onwards all are divisible by 100, we only need to find the last two digits of
2 (f (9)) = 2 (1(1!) + 2(2!) + 3(3!)....... 9(9!))
Now, using the onscreen calculator we can easily calculate the above value and find that the last two digits to be 98. Ans: (98)

3. 7 to 77 - Totally 71 integers. Of these, there are 23 multiples of 3 - i.e., 9, 12, 15, ---- 75. This means that there are (71 – 23) = 48 numbers that are not multiples of 3. To get at least one multiple of 3, we should take all the numbers that are not multiples of 3 and also one number which is a multiple of 3 so that at least one multiple of 3 will be there. Hence, we should take 48 + 1 = 49 numbers. Choice (D)

4. Given that $|x – 3|.(y^2 – 4x + 4) < 0$
Since $|x – 3|$ cannot be less than zero, $y^2 – 4x + 4$ should definitely be less than 0.
$y^2 – 4x + 4 < 0$
$y^2 + 4 < 4x$
$1 + y^2/4 < x$
Since the minimum value of $y^2/4$ is 0, x is definitely greater than 1. Choice (D)

5. Percentage of Kerosene in vessel A
$= \frac{4}{7} \times 100 = 57.14\%$
Percentage of Kerosene in vessel B
$= \frac{8}{19} \times 100 = 42.1\%$
Any mixture formed from these mixtures will always have Kerosene in the range 42.1% to 57.14%.
Considering the choices given, the percentage of kerosene in
Choice (A) = $\frac{13}{30} \times 100 \cong 43.3\%$
Choice (B) = $\frac{11}{20} \times 100 = 55\%$
Choice (C) = $\frac{10}{17} \times 100 \cong 58.8\%$
Choice (D) = $\frac{7}{16} \times 100 = 43.75\%$
The ratio given in Choice (C) is not possible. Choice (C)

6.



Consider ΔABE and ΔACD
∠ABE = ∠ACD (since they are angles subtended by chord DE in the same segment of the circle)
and ∠A is common. Therefore, ∠BEA = ∠CDA)
ΔABE ≈ ΔACD [AAA]
$\therefore \frac{BE}{CD} = \frac{AE}{AD} = \frac{AB}{AC} = \frac{9}{11}$ (given)
Again ΔMBD ≈ ΔMCE [AAA]
[∠DBM = ∠MCE, ∠BDM = ∠CEM and ∠DMB = ∠EMC]
∴ (DM)(MC) = (EM)(MB)
∴ (EM)(MB) = 4(18)
$\therefore \frac{BE}{CD} = \frac{9}{11} \Rightarrow BE = \frac{9}{11}(22) = 18$
∴ EM + MB = 18
$MB + \frac{72}{MB} = 18$ [∵ (EM)(MB) = 72)]
$(MB)^2 – 18(MB) + 72 = 0$
(MB – 6)(MB – 12) = 0
∴ MB = 6 or MB = 12.
As EM < MB (since DM < MC), MB = 12. Choice (D)

7. 2x + 3y + 4z = 33 ---- (1)
4x + 2y + 3z = 29 ---- (2)
3x + 4y + 2z = 28 ---- (3)
(1) + (2) + (3)
⇒ 9 (x + y + z) = 90 ⇒ x + y + z = 10
∴ Choice (A) is consistent.
Accordingly (1) + (2) gives Choice (B)
And (1) – (2) gives Choice (C)
While (2) + (3) – (1) gives 5x + 3y + z = 24
but choice (D) reads as 5x + 3y + z = 23 which is not consistent.

Alternative solution:

Solve the three equations to find out the individual values of x, y and z. Substitute these values and validate the equations given in the options. Only Choice (D) will not be valid. Choice (D)

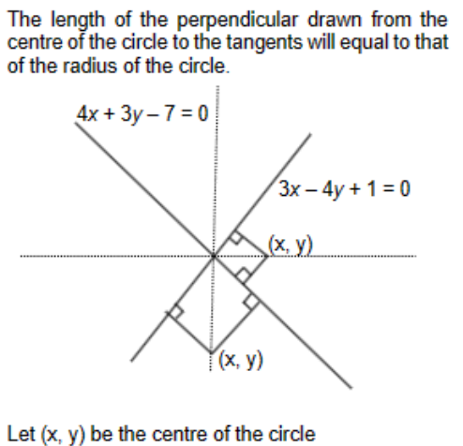
8. $\frac{(4x^{4001} + 5x^{4000})}{(6x^{3999})} = 4$
$\left(\frac{4x^{4001}}{x^{3999}}\right) + \left(\frac{5x^{4000}}{x^{3999}}\right) = 24$
$4x^2 + 5x – 24 = 0$
Product of roots of the equation $= \frac{-24}{4} = -6$ Choice (D)

9. Ratio in which profits are to be divided = (Initial investment of Tarun) × (Number of months) : (Initial investment of Varun) × (Number of months)
= 4 × 8 : 5 × 12 = 8 : 15
Percentage of profit that Tarun will receive
= 8/23 × 100
= 34.78% Choice (A)

10. The first carrier makes a trip every day and the second carrier makes one trip in 2 days and so on. Hence all the trucks will leave on the same day once every N days where N is the LCM of 1, 2, 3, 4 and 5.
⇒ N = 60
∴ If on the 7th of January all the trucks left together then after every 60 days they will leave together (i.e. on the same day). There will be at most 366 – 6 = 360 days including 7th January in that year. In 360 days, there can be at most 6 occasions like that. Ans: (6)

11. 4 male players and 4 female players can be chosen in $6_{C_4} \times 6_{C_4}$ i.e., 225 ways.
Now, we have 4 female players. We need to pair them with the male players. The first female player can be paired in 4 ways, the second in 3 ways, the third in 2 ways and the fourth in 1 way.
Therefore, the total number of selections = 225 × 4! = 5400 Choice (C)

12. The length of the perpendicular drawn from the centre of the circle to the tangents will equal to that of the radius of the circle.



Let (x, y) be the centre of the circle
$\Rightarrow \frac{|4x+3y-7|}{\sqrt{4^2+3^2}} = \frac{|3x-4y+1|}{\sqrt{3^2+4^2}}$
⇒ 4x + 3y – 7 = ±(3x – 4y + 1)
⇒ 4x + 3y – 7 – (3x – 4y + 1) = 0
or
4x + 3y – 7 + (3x – 4y + 1) = 0
⇒ x + 7y – 8 = 0 or 7x – y – 6 = 0
i.e., The centre of the circle lies on either x + 7y – 8 = 0 or 7x – y - 6 = 0
∴ The centre of the circle will definitely satisfy (x + 7y – 8) (7x – y – 6) = 0

Alternative Solution:

The centre of the circle must lie on one of the angle bisectors of the angles formed by the given two

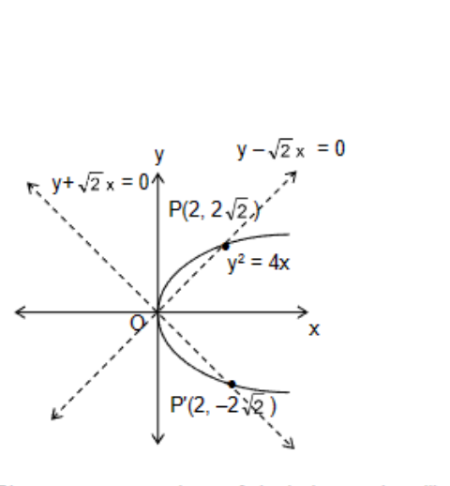
lines. The slopes of the given lines are $\frac{-4}{3}$ and $\frac{3}{4}$, i.e., they are perpendicular to each other.
Hence one of the angle bisectors will be at an angle (say α) of exactly 45° more than the angle, A, made by the line with slope $\left(\frac{3}{4}\right)$, i.e.,
$\tan A = \left(\frac{3}{4}\right)$
$\therefore \tan\alpha = \tan(A + 45°) = \frac{\tan A + 1}{1 - \tan A} = \frac{\frac{3}{4}+1}{1-\frac{3}{4}} = 7$
Hence, slope of the other bisector $= \frac{-1}{7}$
Checking the options (for the slopes of the two lines), only option (B) is possible. Choice (B)

13. Given that the number of bacteria doubles at the end of every twenty minutes.
So, $364 \xrightarrow{20\,min} 728 \xrightarrow{40\,min} 1{,}456 \xrightarrow{60\,min}$
$2{,}912 \xrightarrow{80\,min} 5{,}824 \xrightarrow{100\,min} 10{,}000+$
∴ The count of bacteria will be more than 10,000 in 100 minutes i.e. at 9:40 a.m. Choice (C)

14. Given Ram spent 3 hours and bought 7 popcorns.
Price he pays = 3A + 7B + C
He would have spent ₹300 less had he spent 2.5 hours less and bought 3 popcorns more ⇒ 3A + 7B + C – 0.5A – 10B – C = 300 ⇒ 2.5A – 3B = 300 → (1)
He would have spent ₹300 less had he spent 1.5 hours more and bought 5 popcorns less ⇒ 3A + 7B + C – 4.5A – 2B – C = 300 ⇒ 5B – 1.5A = 300 → (2)
2.5A – 3B = 5B – 1.5A ⇒ 4A = 8B ⇒ A = 2B
The amount spent on every two popcorns is the same as that on staying one hour in the theatre. Hence, if Ram had spent all the money on staying, he could have stayed for $\left(\frac{7}{2}\right)$ = 3.5 hours more, i.e., he could have spent a total of 6.5 hours in the theatre. Ans: (6.5)

15. ₹80 = interest on SI for 1 year
⇒ ₹80 = r% of 1000
⇒ r = 8
Now if 8% of S = 1000
S = ₹12,500 Ans: (12,500)

16. Given curves are:
$y^2 = 2x^2$ and $y^2 = 4x$
i.e., $(y + \sqrt{2}x)(y - \sqrt{2}x) = 0$ and $y^2 = 4x$



Since, y appears only as y^2, both the graphs will be symmetric about the x-axis.
From the figure it is clear that the loops OPO and OP'O are of equal areas (from symmetry about the y-axis).
∴ Ratio of their areas is 1 : 1. Choice (B)

17. L.H.S = 1+3+5+.....(2n – 1), i.e., the sum of the first 'n' odd numbers, which is equal to n^2.
Since m = 25, R.H.S = 1+25.26.27.28 = 491401
Hence, n^2 = 491401 and n = $\sqrt{491401}$ = 701
(The on-screen calculator be used for the calculations)

Alternative Solution:

m (m+1) (m+2) (m+3)
= m (m+3) (m+1) (m+2)
= $[m^2 + 3m]\,[m^2+3m+2]$
= $[m^2 + 3m+1–1]\,[m^2+3m+1+1]$
= $(m^2+3m+1)^2 – 1$
⇒ 1+m (m+1) (m+2) (m+3) = $(m^2+3m+1)^2$
Given m = 25, we get $[25^2+3(25)+1]^2 = 701^2$.
Now 1+ 3 +5 +.... + (2n – 1) = n^2
Now, $n^2 = 701^2$
⇒ n = 701 Ans: (701)

18. Let the number be x. The student first added 6 to the number and then divided it by 7. Hence $\frac{(x+6)}{7}$ = 25
x = 25 × 7 – 6 = 169
Correct answer should have been $\frac{169}{7} + 6 = \frac{211}{7}$ Choice (A)

19. Let the number of children who are 8 years old and 7 years old be a and b respectively.
Given that 8a + 7b = 70.
The only values of a and b which satisfy the above equation are a = 7 and b = 2. Hence, there are seven 8-year-olds and two 7-year-olds.
A team should be selected from among these children such that the total sum of their ages is 54.
i.e 8x + 7y = 54
x can be a maximum of 7 and y can be a maximum of 2.

Based on the above, the only values of x and y that satisfy the given equation are x = 5 and y = 2.
This means that five 8-year-olds and both the 7-year-olds must be selected.
The number of ways of doing so will be $^7C_5 \times {}^2C_2$
= 21 ways. Ans: (21)

20. For any triangle, the sides are inversely proportional to corresponding the altitudes.
From statement I,
The ratio of the sides of the triangle ABC
$= \frac{1}{2} : \frac{1}{5} : \frac{1}{6}$
= 30 : 12 : 10
This is not possible as the sum of any two sides in a triangle must be greater than the third side and 10 + 12 ≯ 30.
From statement II,
The ratio of the sides of the triangle DEF
$= \frac{1}{3} : \frac{1}{4} : \frac{1}{8}$
= 32 : 24 : 12
This is possible as it satisfies the triangle inequality Choice (B)

21. Let the shares of land be a, ar and ar^2 (r > 1). Now, given $a(r^2 – 1) = 385$ and $a(r + 1) = 770$
$\Rightarrow \frac{a(r^2-1)}{a(r+1)} = \frac{385}{770}$
⇒ r – 1 = 0.5
⇒ r = 1.5 and a = 308
Hence A = 308 (1 + 1.5 + 2.25) = 1463. Choice (C)

22. Let efficiency (i.e., work done per day) of Q = 1
⇒ efficiency of P = 1.3
P alone takes 46 days.
That is, it takes 46 days to complete the work with an efficiency of 1.3.
Hence, it will take $46 \times \frac{(1.3)}{(1+1.3)}$ = 26 days to complete the work when both P and Q work together. Ans: (26)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	18
Easy	1, 8, 9, 13, 15, 22
Medium	3, 5, 7, 10, 11, 16, 17, 20, 21
Difficult	2, 4, 6, 12, 14, 19
Very Difficult	–

Done
Tip: Scroll slowly for the best results